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Lab 4: Implementation and Comparison of Agile Project Management
Tools – JIRA, Asana, Trello, ClickUp, and Zoho

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OBJECTIVE:

This lab centers on setting up and comparing several Agile project management tools, namely JIRA, Asana, Trello, ClickUp, and Zoho.

AIM :

To explore, evaluate, and configure a set of Agile project management tools, namely JIRA, Asana, Trello, ClickUp, and Zoho Projects, for the purposes of sprint planning, task management, and progress tracking within Agile workflows.

THEORY :

Agile project management tools offer digital Kanban boards, sprint planning utilities, issue trackers, and collaboration features that support both Scrum and Kanban methodologies.

- JIRA (Atlassian): the industry-standard choice for software teams, offering Scrum boards, Kanban boards, epics, stories, sprints, and backlog management.
- Asana: a task and project management tool offering timeline (Gantt), board, and list views, suitable for both technical and non-technical teams alike.
- Trello: a straightforward Kanban board tool built around cards and columns, well suited to small teams with simple workflows.
- ClickUp: an all-in-one productivity platform that merges tasks, documents, goals, and time tracking, offering several views such as list, board, Gantt, and calendar.
- Zoho Projects: provides task management, Gantt charts, time logging, and integrates smoothly with other applications in the Zoho suite.

OBJECTIVES :

- Set up a project and board within JIRA, Trello, and ClickUp.
- Define a sprint that includes user stories and their associated tasks.
- Move tasks between workflow stages, such as To Do, In Progress, and Done.

- Compare the feature sets offered across the different tools.

ALGORITHM :

1. Register an account on each of JIRA, Trello, Asana, ClickUp, and Zoho.
2. Create a new project titled 'Agile Lab Project' within each tool.
3. Add a board containing the columns To Do, In Progress, Review, and Done.
4. Draft user stories and break each of them down into tasks with assigned due dates.
5. Move the tasks progressively through the various board stages.
6. Compare sprint velocity, reporting capability, and overall ease of use among the tools.

CODE :

JIRA – Creating an issue via REST API:

```
curl -X POST https://your-domain.atlassian.net/rest/api/3/issue \
-H "Content-Type: application/json" \
-u email@example.com:API_TOKEN \
-d '{ "fields": { "project": { "key": "LAB" },
      "summary": "Implement login feature",
      "issuetype": { "name": "Story" } } }'
```

Trello – Adding a card via API:

```
curl -X POST "https://api.trello.com/1/cards" \
-d "name=Implement+Login&idList=LIST_ID&key=API_KEY&token=TOKEN"
```

OUTPUT :

- JIRA: a sprint board was set up containing 5 user stories distributed across the workflow columns.
- Trello: a Kanban board was configured, with cards successfully moved between lists.
- Asana: the timeline view clearly presented the dependencies between tasks.
- ClickUp: the dashboard displayed the task completion percentage together with time estimates.
- Zoho: a Gantt chart was generated automatically based on the task due dates.

RESULT :

All five Agile project management tools were configured successfully and used to manage a sample sprint.

CONCLUSION :

Different Agile PM tools cater to different team needs: JIRA is well suited to large software teams, Trello suits small teams that value simplicity, ClickUp and Asana strike a balance between functionality and usability, and Zoho integrates neatly within a broader business software suite.